ma4261_hw3

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August 2024

1 Question 1

Let $p_{Y|X}$ be a ternary channel with input symbol $\mathcal{X} = \mathcal{Y} = \{0, 1, 2\}$ given by

$$P_{Y|X}(y|x) = \begin{cases} 1-\epsilon & \text{if } x = y \neq 1 \\ \frac{1-\epsilon}{2} & \text{if } x = 1 \text{ and } y \neq 1 \\ \epsilon & \text{if } y = 1 \\ 0 & \text{else} \end{cases}$$

(a) Draw a diagram with the transition probabilities for this channel

(b) Observe any symmetries of this channel. List the permutations σ on $\mathcal{X}$ for which there exists a permutation σ' on $\mathcal{Y}$ such that $p_{Y|X}(y|x) = p_{Y|X}(\sigma'(y)|\sigma(x))$ for all x, y

(c) We are interested in the "information channel capacity" $C(p_{Y|X}) = \max_{p_X} I(p_X, P_{Y|X})$. Using the symmetry of the channel, prove that the capacity-achieving input distribution $C(p_{Y|X})$ can be chosen of the following form:

$$p_X^*(x) = \begin{cases} \frac{1-q}{2} & \text{if } x \neq 1 \\ q & \text{if } x = 1 \end{cases}$$

for some $q \in [0, 1]$ still to be determined.

(d) Using the simplification established in (c) to compute $I(p_X, p_{Y|X})$ as a function of q. Determine $C(p_{Y|X})$ and the value q^* that achieves the maximum.

(e) If we fix $q = q^*$, the channel effectively reduces to a well-know channel. Which one?

1.1 a

Figure 1

1.2 b

Symmetry of the channel

$$\sigma = \sigma' = (2, 1, 0) = \{(0 \mapsto 2), (2 \mapsto 0)\}$$

1.3 c

As $p_X \mapsto I(p_X, p_{Y|X})$ is concave on a convex, compact set, it is either a constant or have exactly one maximum. In both case, if $(0, 1, 2) \mapsto (a, b, c)$ maximizes $I(p_X, p_{Y|X})$, then $(0, 1, 2) \mapsto (c, b, a)$ also maximizes $I(p_X, p_{Y|X})$. Therefore, given p_X^*, we must have $p_X^*(0) = p_X^*(2)$. Therefore, p_X^* is of the form

$$p_X^*(x) = \begin{cases} \frac{1-q}{2} & \text{if } x \neq 1 \\ q & \text{if } x = 1 \end{cases}$$

for $q \in [0, 1]$

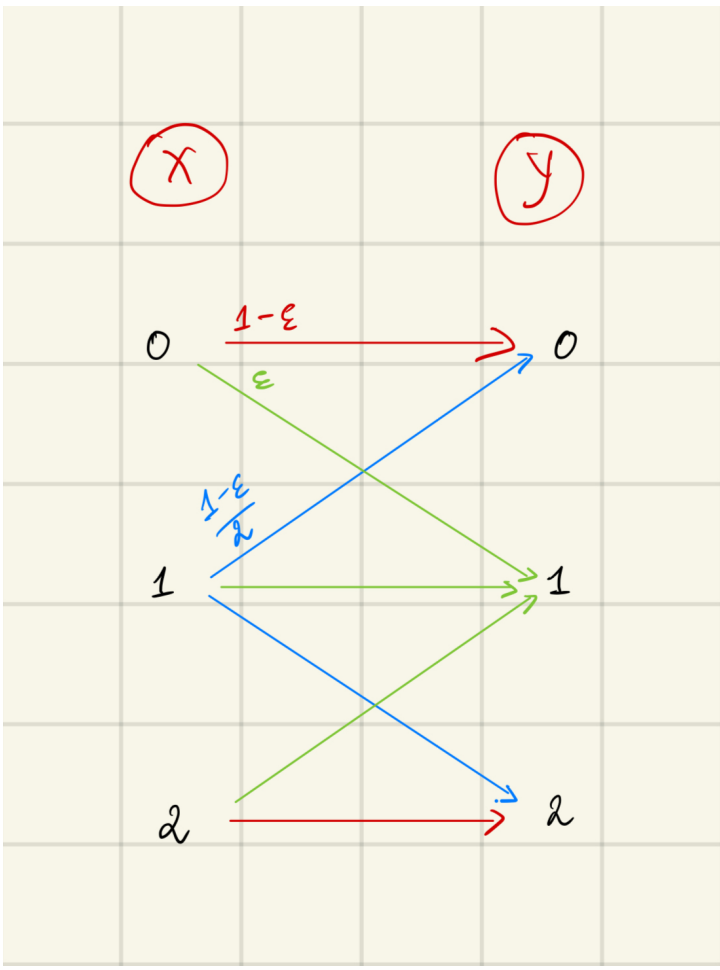

Figure 1: channel 1a

1.4 d

Let $p_X = p_X^*$. Calculate $p_Y(y)$

$$
\begin{aligned}
p_Y(y) &= \sum_{x \in \mathcal{X}} p_{Y|X}(y|x) p_X(x) \\
&= p_{Y|X}(y|0)p_X(0) + p_{Y|X}(y|1)p_X(1) + p_{Y|X}(y|2)p_X(2) \\
&= p_{Y|X}(y|0)p_X(0) + p_{Y|X}(y|1)p_X(1) \\
&= \frac{1-q}{2} p_{Y|X}(y|0) + q p_{Y|X}(y|1) + \frac{1-q}{2} p_{Y|X}(y|2)
\end{aligned}
$$

Then

$$
\begin{aligned}
p_Y(0) &= \frac{1-q}{2}(1-\epsilon) + q\frac{1-\epsilon}{2} = \frac{1-\epsilon}{2} \\
p_Y(1) &= \frac{1-q}{2}\epsilon + q\epsilon + \frac{1-q}{2}\epsilon = \epsilon \\
p_Y(2) &= q\frac{1-\epsilon}{2} + \frac{1-q}{2}(1-\epsilon) = \frac{1-\epsilon}{2}
\end{aligned}
$$

We have

$$\begin{aligned}
H(Y) &= \sum_{y\in\mathcal{Y}} p_Y(y)\log\frac{1}{p_Y(y)} \\
&= p_Y(0)\log\frac{1}{p_Y(0)} + p_Y(1)\log\frac{1}{p_Y(1)} + p_Y(2)\log\frac{1}{p_Y(2)} \\
&= \frac{1-\epsilon}{2}\log\frac{2}{1-\epsilon} + \epsilon\log\frac{1}{\epsilon} + \frac{1-\epsilon}{2}\log\frac{2}{1-\epsilon} \\
&= (1-\epsilon)\log\frac{2}{1-\epsilon} + \epsilon\log\frac{1}{\epsilon}
\end{aligned}$$

Let $H_b(\epsilon)$ denote the entropy of the distribution $[\epsilon, 1-\epsilon]$ and $H_t(1/3)$ denote the entropy of the distribution $[1/3, 1/3, 1/3]$

$$\begin{aligned}
H(Y|X) &= \sum_{x\in\mathcal{X}} p_X(x)H(Y|X=x) \\
&= p_X(0)H(Y|X=0) + p_X(1)H(Y|X=1) + p_X(2)H(Y|X=2) \\
&= p_X(0)H_b(\epsilon) + p_X(1)H_t(1/3) + p_X(2)H_b(\epsilon) \\
&= (1-q)H_b(\epsilon) + qH_t(1/3)
\end{aligned}$$

Note that, $H(Y|X)$ is a convex combination of $(H_b(\epsilon), H_t(1/3))$ that achieves minimum at $\min\{H_b(\epsilon), H_t(1/3)\}$. As $H_b(\epsilon) \le 1 < \log 3 = H_t(1/3)$, therefore, $H(Y|X)$ is minimized at $q^* = 0$. Since, $H(Y)$ is independent of q, then $I(X;Y) = H(Y) - H(Y|X)$ is maximized at $q^* = 0$, and the maximum value is

$$\begin{aligned}
C(p_{Y|X}) &= H(Y) - H_b(\epsilon) \\
&= (1-\epsilon)\log\frac{2}{1-\epsilon} + \epsilon\log\frac{1}{\epsilon} - (1-\epsilon)\log\frac{1}{1-\epsilon} - \epsilon\log\frac{1}{\epsilon} \\
&= 1-\epsilon
\end{aligned}$$

1.5 e

If we fix $q = q^* = 0$, the channel reduces to binary erasure channel (BEC) of erasing probability ϵ

2 Question 2

Consider the conditional distribution $P_{Y^m|X}$ that stochastically maps $X \in [0,1]$ to $Y^m \in \{0,1\}^m$ where given $x \in [0,1]$, Y^m is i.i.d Bernoulli(x). define the m-information channel capacity

$$C(m) = \max_{p_X} I(X; Y^m)$$

We aim to prove, in this question that

$$\frac{1}{2}\log m + O(1) \le C(m) \le \log m + O(1)$$

where $O(1)$ is a term that is bounded as $m \to \infty$

(a) Let $S = \sum_{i=1}^m Y_i$. Prove that $C(m) = \max_{p_X} I(X, S)$

(b) Let $\tilde{C}(p_{Y|X}) = \max_{p_X} I(p_X, p_{Y|X})$ be the usual information channel capacity. Prove that

$$\tilde{C}(p_{Y|X}) = \max_{p_X}\min_{q_Y} D(p_{Y|X}||q_Y|p_X) = \min_{q_Y}\max_{p_X} D(p_{Y|X}||q_Y|p_X)$$

This is known as the saddle point property of the capacity function.

(c) Show using the previous part that

$$C(m) = \min_{q_S}\sup_{x\in[0,1]} D(Binomial(m,x)||q_S)$$

where q_S is a distribution supported on $\{0, 1, .., m\}$

(d) Choose uniform q_S to show that

$$C(m) \leq \log m + O(1)$$

as $m \to \infty$

(e) Choose uniform p_X to show that for all $\epsilon > 0$, there exists $m_0(\epsilon)$ such that for all $m > m_0(\epsilon)$

$$C(m) \geq \frac{1-\epsilon}{2} \log m$$

Given a Markov chain

2.1 a

$$X \longrightarrow Y^m \longrightarrow S(Y)$$

As sample mean is a sufficient statistic for Bernoulli RVs, $I(X, Y^m) = I(X, S(Y))$, therefore

$$C(m) = \max_{p_X} I(X, Y^m) = \max_{p_X} I(X, S)$$

2.2 b

We have

$$\begin{aligned}
I(p_X, p_{Y|X}) &= \sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} p_{XY}(x,y) \log \frac{p_{XY}(x,y)}{p_X(x)p_Y(y)} \\
&= \sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} p_X(x)p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{p_Y(y)} \\
&= \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{p_Y(y)}
\end{aligned}$$

Let $q^*_{Y|X}(y|x) = p_Y(y)$ be a conditional probability that is independent of X

$$I(p_X, p_{Y|X}) = \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} = D(p_{Y|X}||q^*_{Y|X}|p_X)$$

Moreover, for any distribution $q_{Y|X}(y|x)$

$$\begin{aligned}
I(p_X, p_{Y|X}) &= \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} \\
&= \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{q_{Y|X}(y|x)} \frac{q_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} \\
&= \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{p_{Y|X}(y|x)}{q_{Y|X}(y|x)} + \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{q_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} \\
&= D(p_{Y|X}||q_{Y|X}|p_X) + \sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p_{Y|X}(y|x) \log \frac{q_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} \\
&= D(p_{Y|X}||q_{Y|X}|p_X) + \sum_{y\in\mathcal{Y}} p_Y(y) \log \frac{q_{Y|X}(y|x)}{q^*_{Y|X}(y|x)} \\
&= D(p_{Y|X}||q_{Y|X}|p_X) - \sum_{y\in\mathcal{Y}} q^*_{Y|X}(y|x) \log \frac{q^*_{Y|X}(y|x)}{q_{Y|X}(y|x)} \\
&= D(p_{Y|X}||q_{Y|X}|p_X) - D(q^*_{Y|X}(\cdot|x)||q_{Y|X}(\cdot|x)) \\
&\leq D(p_{Y|X}||q_{Y|X}|p_X)
\end{aligned}$$

Therefore, we have

$$D(p_{Y|X}||q^*_{Y|X}|p_X) = I(p_X, p_{Y|X}) \leq D(p_{Y|X}||q_{Y|X}|p_X)$$

Then,

$$I(p_X, p_{Y|X}) = \min_{q_{Y|X}} D(p_{Y|X}||q_{Y|X}|p_X)$$

As $\min_{q_{Y|X}} D(p_{Y|X}||q_{Y|X}|p_X)$ reach its minimum at $q^*_{Y|X} = p_Y$ independent of X, therefore,

$$I(p_X, p_{Y|X}) = \min_{q_Y} D(p_{Y|X}||q_Y|p_X)$$

Hence,

$$\tilde{C}(p_X, p_{Y|X}) = \max_{p_X} \min_{q_{Y|X}} D(p_{Y|X}||q_Y|p_X)$$

By minimax theorem,

$$\tilde{C}(p_X, p_{Y|X}) = \max_{p_X} \min_{q_Y} D(p_{Y|X}||q_Y|p_X) = \min_{q_Y} \max_{p_X} D(p_{Y|X}||q_Y|p_X)$$

2.3 c

We have

$$\begin{aligned} C(m) &= \max_{p_X} I(X, S) \\ &= \min_{q_{S|X}} \max_{p_X} D(p_{S|X}||q_{S|X}|p_X) \\ &= \min_{q_{S|X}} \max_{p_X} \int_{x\in[0,1]} p_X(x) D(p_{S|X}(\cdot|x)||q_{S|X}(\cdot|x))dx \end{aligned}$$

Note that, the integral $\int_{x\in[0,1]} p_X(x)D(p_{S|X}(\cdot|x)||q_{S|X}(\cdot|x))dx$ is maximized when p_X concentrate at one point such that it maximizes $D(p_{S|X}(\cdot|x)||q_{S|X}(\cdot|x))$, therefore

$$C(m) = \min_{q_{S|X}} \sup_{x\in[0,1]} D(p_{S|X}(\cdot|x)||q_{S|X}(\cdot|x))$$

Given any $x \in [0,1]$, $p_{S|X}(\cdot|x) = Binomial(m, x)$ and from previous derivations, $q_{S|X}$ independent of X gives the maximum value for $D(p_{S|X}(\cdot|x)||q_{S|X}(\cdot|x))$, then

$$C(m) = \min_{q_S} \sup_{x\in[0,1]} D(Binomial(m, x)||q_S)$$

2.4 d

Let q_S be uniform

$$\begin{aligned} D(Binomial(m, x)||q_S) &= \sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{n-k} \log \frac{\binom{m}{k} x^k (1-x)^{n-k}}{1/m} \\ &= \log m - H(Binomial(m, x)) \leq \log m \end{aligned}$$

Therefore,

$$C(m) \leq \log m$$

for all $m \in \mathbb{N}$. Another proof is that $I(X, S) \leq H(S) \leq \log m$

2.5 e

Lemma 1. *For any integer-valued random variable X with variance σ^2, then*

$$H(X) \leq \frac{1}{2} \log \left[2\pi e \left(\sigma^2 + \frac{1}{12} \right) \right]$$

Moreover, if X is $Binomial(n, p)$, then $\sigma^2 = np(1-p)$, we have

$$H(X) \leq \frac{1}{2} \log(np(1-p))$$

Main Proof:

$$
\begin{aligned}
D(Binomial(m,x)||q_S) &= \sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{n-k} \log \frac{\binom{m}{k} x^k (1-x)^{n-k}}{q_S(k)} \\
&= -H(Binomial(m,x)) + \sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{n-k} \log \frac{1}{q_S(k)} \\
&\geq -H(Binomial(m,x)) + \log \left[\sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{n-k} \frac{1}{q_S(k)} \right] && \text{(Jensen)} \\
&\geq -H(Binomial(m,x)) + \log \left[\sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{n-k} \right] && (\frac{1}{q_S(k)} \geq 1)
\end{aligned}
$$

TODO - right term is cross entropy

3 Question 3

The problem deals with the converse proof of Shannon's Channel Coding Theorem under the criterion of an "average bit error probability" of decoding

Consider a message $W = (W_1, ..., W_{nR})$ uniformly distributed over $\{0,1\}^{nR}$, i.e., W consists of nR (treat this as an integer) independent bits $W_1, ..., W_{nR}$. Consider a sequence of $(2nR, n)$-codes for the channel with encoder f_n and decoder ϕ_n. Recall that the average error probability is

$$\lambda_{ave}(f_n, \phi_n) = Pr(W \neq \hat{W})$$

where $\hat{W} = \phi_n(Y^n)$. Denote by $\hat{W}_i$ the estimate of the i-th message bit W_i and let

$$b_i(f_n, \phi_n) = Pr(W_i \neq \hat{W}_i)$$

for $1 \leq i \leq nR$. Furthermore, let the average bit error probability be defined by

$$b_{ave}(f_n, \phi_n) = \frac{1}{nR} \sum_{i=1}^{nR} b_i(f_n, \phi_n)$$

(a) Show that $b_{ave}(f_n, \phi_n) \leq \lambda_{ave}(f_n, \phi_n)$

(b) In class we showed the weak converse to Shannon's channel coding theorem which says that under $\lambda_{ave}(f_n, \phi_n) \to 0$, $R \leq C = \max_{p_X} I(p_X, p_{Y|X})$. Show that even if $b_{ave}(f_n, \phi_n) \to 0$, the rate satisfies

$$R \leq C = \max_{p_X} I(p_X, p_{Y|X})$$

3.1 a

We have

$$
\begin{aligned}
\lambda_{ave}(f_n, \phi_n) &= Pr(W \neq \hat{W}) \\
&= 1 - Pr(W_1 = \hat{W}_1, W_2 = \hat{W}_2, ..., W_{nR} = \hat{W}_{nR}) \\
&\geq 1 - \min_{i \in [nR]} Pr(W_i = \hat{W}_i) && \text{(monotone)} \\
&\geq 1 - \frac{1}{nR} \sum_{i=1}^{nR} Pr(W_i = \hat{W}_i) && \text{(minimum is less than average)} \\
&= \frac{1}{nR} \sum_{i=1}^{nR} \Big(1 - Pr(W_i = \hat{W}_i)\Big) \\
&= \frac{1}{nR} \sum_{i=1}^{nR} Pr(W_i \neq \hat{W}_i) \\
&= b_{ave}(f_n, \phi_n)
\end{aligned}
$$

3.2 b

Consider the Markov chain

$$(W_1, W_2, ..., W_{nR}) - X^n - Y^n - (\hat{W}_1, \hat{W}_2, ..., \hat{W}_{nR})$$

Since W is uniformly distributed on $\{0,1\}^{nR}$, bits are independent, then

$$\sum_{i=1}^{nR} I(W_i; \hat{W}_i) \leq I(W; \hat{W})$$

$$\begin{aligned} nC &\geq I(X^n; Y^n) \\ &\geq I(W, \hat{W}) && \text{(DPI)} \\ &\geq \sum_{i=1}^{nR} I(W_i; \hat{W}_i) \end{aligned}$$

Let $p_e^i = P(W_i \neq \hat{W}_i)$, note that since P_{W_i} is uniform on $\{0,1\}$, then

$$\begin{aligned} I(W_i; \hat{W}_i) &\geq d_b(P_{W_i\hat{W}_i}(W_i = \hat{W}_i), (P_{W_i} \times P_{\hat{W}_i})(W_i = \hat{W}_i)) \\ &= d_b(1 - p_e^i, 1/2) \\ &= (1 - p_e^i) \log \frac{1 - p_e^i}{1/2} + p_e^i \log \frac{p_e^i}{1/2} \\ &= 1 - \log(1 - p_e^i) \\ &\geq 1 - \frac{1}{\ln 2} p_e^i && \text{(for } p_e^i \text{ small and positive)} \end{aligned}$$

Then

$$nC \geq \sum_{i=1}^{nR} (1 - \frac{1}{\ln 2} p_e^i)$$

Then

$$C \geq R - R\frac{1}{\ln 2} b_{ave}(f_n, \phi_n)$$

As $b_{ave}(f_n, \phi_n) \to 0$, we must have $R \leq C$

4 Question 4

[Deterministic Channel] Consider a memoryless channel that takes pairs of bits as input and produces two bits as output as follows: $00 \mapsto 01, 01 \mapsto 10, 10 \mapsto 11, 11 \mapsto 00$. Let (X_1, X_2) denote the two input bits and (Y_1, Y_2) the two output bits

(a) Calculate the mutual information $I(X_1, X_2; Y_1, Y_2)$ for a given joint PMF of four pairs of input bits. You can express your answer in terms of

$$\begin{aligned} p_{00} &= Pr(X_1 = 0, X_2 = 0) \\ p_{10} &= Pr(X_1 = 1, X_2 = 0) \\ p_{01} &= Pr(X_1 = 0, X_2 = 1) \\ p_{11} &= Pr(X_1 = 1, X_2 = 1) \end{aligned}$$

(b) Show that the capacity is 2 and indicate the units of capacity

(c) Show that, surprisingly $I(X_1, Y_1) = 0$ for the capacity-maximizing distribution of the input you derived in part (b), that is, information is only transferred by considering both bits

4.1 a

$$
\begin{aligned}
I(X_1, X_2; Y_1, Y_2) &= H(Y_1, Y_2) - H(Y_1, Y_2 | X_1, X_2) \\
&= H(Y_1, Y_2) && \text{(deterministic)} \\
&= H(X_1, X_2) && \text{(injective)} \\
&= \sum_{x_1 x_2 \in \{00,01,10,11\}} p_{x_1 x_2} \log \frac{1}{p_{x_1 x_2}}
\end{aligned}
$$

4.2 b

$I(X_1, X_2; Y_1, Y_2)$ is maximized when $X_1 X_2$ is uniform, that is

$$C = \log \frac{1}{1/4} = 2$$

4.3 c

$$
\begin{aligned}
I(X_1, Y_1) &= \sum_{(x,y) \in \{0,1\}^2} Pr(X_1 = x, Y_1 = y) \log \frac{Pr(X_1 = x, Y_1 = y)}{Pr(X_1 = x) Pr(Y_1 = y)} \\
&= \sum_{(x,y) \in \{0,1\}^2} Pr(X_1 = x, Y_1 = y) \log \frac{Pr(Y_1 = y | X_1 = x)}{Pr(Y_1 = y)}
\end{aligned}
$$

If X is uniform, for every pair $(x, y) \in \{0, 1\}^2$, we always have

$$Pr(Y_1 = y | X_1 = x) = Pr(Y_1 = y) = \frac{1}{2}$$

Therefore,

$$I(X_1, Y_1) = 0$$